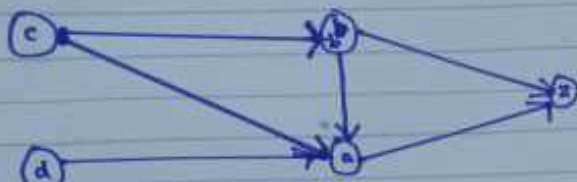


①

Problem 01 COMPUTATIONAL GRAPHS



$$\rightarrow a = 3d^3 + 2c + b^2$$

$$\rightarrow b = 2c^2$$

$$\rightarrow z = 3a + 2b^2$$

$$\rightarrow \text{where } c=2 \text{ and } d=5$$

Perform the forward pass and compute the values of z, b, a

From the graph:

$$c \rightarrow b$$

then b and d contribute to a , and finally a and b contribute to z .

So we will calculate:

$$b \rightarrow a \rightarrow z$$

① Calculate b :

We have:

$$b = 2c^2$$

Since:

$$c = 2$$

Substitute:

$$b = 2(2)^2$$

$$b = 2(4)$$

$$b = 8$$

② Calculate a :

We have:

$$a = 3d^3 + 2c + b^2$$

We already know: $d=5, c=2, b=8$

$$d=5, c=2, b=8$$

Substitute these values:

$$a = 3(5)^3 + 2(2) + (8)^2$$

Calculate each part

$$5^2 = 125$$

$$3(125) = 375$$

$$2(2) = 4$$

$$8^2 = 64$$

Therefore:

$$a = 375 + 4 + 64$$

$$a = 443$$

③ Calculate Z:

We have:

$$Z = 3a + 2b$$

Since:

$$a = 443, \quad b = 8$$

substitute

$$Z = 3(443) + 2(8)$$

$$Z = 1329 + 16$$

Hence:

$$Z = 1345$$

→ For the forward pass:

$$b^2 = 8^2 = 64$$

$$a = 443$$

$$Z = 1345$$

PART 2

Using chain rule compute the value of $\frac{dz}{dc}$. Remember direct substitution of values will lead to 0 marks.

→ We have:

$$a = 3d^3 + 2c + b^2$$

$$b = 12c^2 + 2$$

$$Z = 3a + 2b$$

We want:

$$\frac{dz}{dc}$$

$$dc$$

① Understands how c affects z : $\frac{\partial z}{\partial c}$

If we look at the diagram:

$\rightarrow c$ affects b :

$$c \rightarrow b$$

b then affects a :

$$b \rightarrow a$$

And both a and b affect z :

$$a \rightarrow z$$

$$b \rightarrow z$$

Therefore, there are more than 1 path from c to z .

Path 1:

$$c \rightarrow b \rightarrow z$$

Path 2:

$$c \rightarrow b \rightarrow a \rightarrow z$$

There is also a direct path:

$$c \rightarrow a \rightarrow z$$

So we need to account for all three paths.

② Apply the chain rule:

For z , which depends on both a and b :

$$\frac{dz}{dc} = \frac{\partial z}{\partial a} \frac{da}{dc} + \frac{\partial z}{\partial b} \frac{db}{dc}$$

This is the key chain-rule equation

③ Calculate $\frac{\partial z}{\partial a}$

We have:

$$z = 3a + 2b$$

Differentiate with respect to a :

$$\boxed{\frac{\partial z}{\partial a} = 3}$$

④ Calculate $\frac{\partial z}{\partial b}$: ::

Again :

$$z = 3a + 2b$$

Differentiate with respect to b :

$$\frac{\partial z}{\partial b} = 2$$

⑤ Calculate $\frac{db}{dc}$:

We have :

$$b = 2c^2$$

Differentiate :

$$\frac{db}{dc} = 4c$$

Now, because we are eventually evaluating the derivatives at the given points $a=2, c=1$:

$$\frac{db}{dc} = 4(1)$$

$$\frac{db}{dc} = 4$$

(2)

⑥ Calculate $\frac{da}{dc}$:

We have:

$$A = 3d^2 + 2c + b^2$$

There are 2 ways c affects a :

1. Directly through $2c$
2. Indirectly through b^2

Therefore:

$$\frac{da}{dc} = \frac{\partial A}{\partial c} + \frac{\partial A}{\partial b} \frac{db}{dc}$$

Now calculate each component

Direct effect^t of c :

$$\frac{\partial A}{\partial c} = 2$$

Effect through b :

Since:

$$A = 3d^2 + 2c + b^2$$

we have:

$$\frac{\partial A}{\partial b} = 2b$$

We already found from the forward pass that $b = 8$

Therefore:

$$\frac{\partial A}{\partial b} = 2(8) = 16$$

And:

$$\frac{db}{dc} = 28$$

So:

$$\frac{da}{dc} = 2 + (16)(8)$$

$$= 2 + 128$$

$$\boxed{\frac{da}{dc} = 130}$$

⑦ Put Everything Together:

We started with:

$$\frac{dz}{dc} = \frac{\partial z}{\partial a} \frac{da}{dc} + \frac{\partial z}{\partial b} \frac{db}{dc}$$

Now we have:

$$\frac{\partial z}{\partial a} = 3$$

$$\frac{\partial z}{\partial b}$$

$$da = 130$$

$$db$$

$$\frac{\partial z}{\partial a} = 2$$

$$\frac{\partial z}{\partial b}$$

$$db = 8$$

$$dc$$

Therefore $\frac{dz}{dc} = (3)(130) + 2(8)$

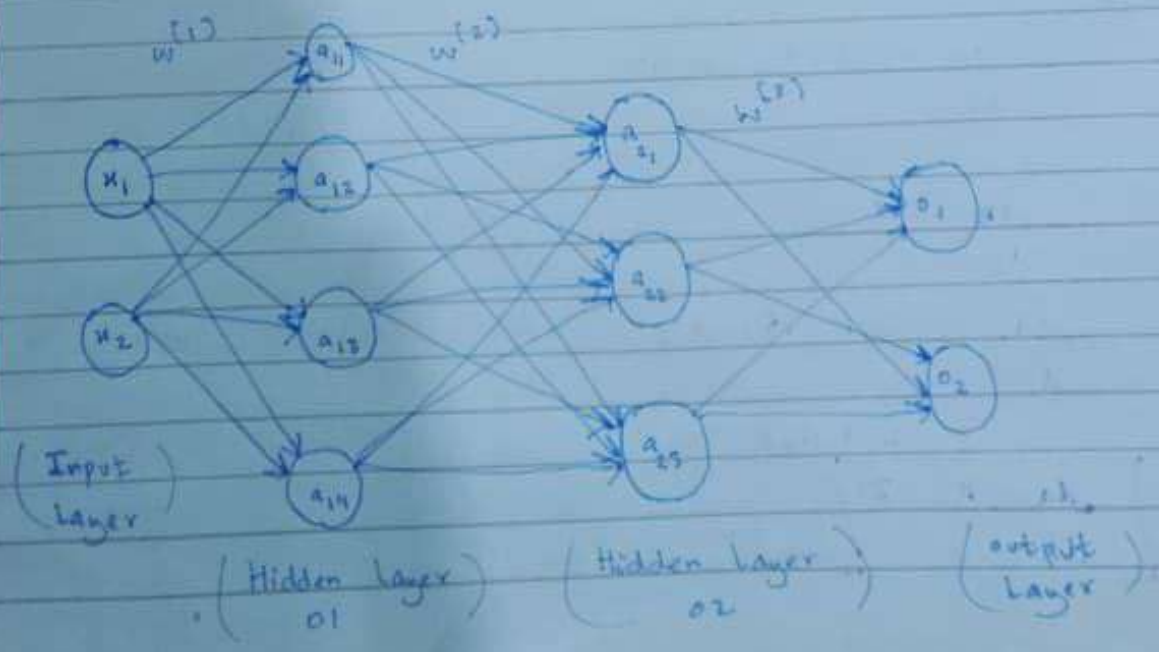
$$= 390 + 16$$

$$\frac{dz}{dc} = 406$$

PROBLEM 02

"BACK PROPAGATION"

Consider the neural network diagram given below:



PROBLEM 2 $\rightarrow$ PART (A) : First Hidden Layer (1-1a) (3)

The assignment gives

$$u_1 = 0.5, \quad u_2 = 0.3$$

All the biases are 0.

Hidden layers use tanh.

First hidden layer has 4 nodes

$$a_{11}, a_{12}, a_{13}, a_{14}$$

Step (1) : Calculate a_{11} :

The weights from u_1, u_2 to a_{11} are

$$w_{11}^{(1)} = 0.9 \quad z_1$$

$$w_{21}^{(1)} = -0.79$$

So the weighted sum entering a_{11} is :

$$z_{11} = u_1 w_{11}^{(1)} + u_2 w_{21}^{(1)}$$

Substitute:

$$\begin{aligned} z_{11} &= (0.5)(0.9) + (0.3)(-0.79) \\ &= 0.45 - 0.237 \\ &= 0.213 \end{aligned} \quad (9)$$

Since, this is a hidden layer node, apply tanh:

$$a_{11} = \tanh(0.213)$$

$$a_{11} = 0.20985$$

Step (2) : Calculate a_{12} :

The relevant weights are:

$$w_{12}^{(1)} = 0.75$$

$$w_{22}^{(1)} = -0.03$$

Therefore:

$$z_{12} = u_1 w_{12}^{(1)} + u_2 w_{22}^{(1)}$$

$$= (0.5)(0.75) + (0.3)(-0.03)$$

$$= 0.375 - 0.009$$

$$= 0.366$$

Apply tanh:

$$a_{12} = \tanh(0.366) \rightarrow a_{12} = 0.35048$$

3

PROBLEM 02 → PART A → SECOND HIDDEN LAYER

① CALCULATE A_{21} :

The weights entering A_{21} are:

$$w_{11}^{(2)} = -0.58$$

$$w_{12}^{(2)} = 0.23$$

$$w_{13}^{(2)} = -0.26$$

$$w_{14}^{(2)} = -0.09$$

So, first calculate the weighted sum:

$$Z_{21} = a_1 w_{11}^{(2)} + a_2 w_{12}^{(2)} + a_3 w_{13}^{(2)} + a_4 w_{14}^{(2)}$$

Substitute:

$$Z_{21} = (0.28484)(-0.58) + (0.7549)(0.23) + (-0.29983)(-0.26) + (0.16955)(-0.09)$$

Calculate:

$$Z_{21} = -0.12171 + 0.18067 + 0.07795 - 0.01524$$

$$Z_{21} = -0.06758$$

Apply tanh:

$$A_{21} = \tanh(-0.06758)$$

$$A_{21} = -0.06747$$

② CALCULATE A_{22} :

The weights entering A_{22} are:

$$w_{21}^{(2)} = 1.0$$

$$w_{22}^{(2)} = 0.45$$

$$w_{23}^{(2)} = -0.65$$

$$w_{24}^{(2)} = -0.57$$

③ Calculate a_{13} :
The weights are:

$$w_{13}^{(2)} = -0.15$$

$$w_{23}^{(2)} = 0.10$$

Therefore:

$$z_{13} = x_1 w_{13}^{(2)} + x_2 w_{23}^{(2)}$$

$$= (0.5)(-0.15) + (0.3)(0.10)$$

$$= -0.075 + 0.03$$

$$= -0.045$$

Apply Tanh:

$$a_{13} = \tanh(-0.045)$$

$$a_{13} = -0.04497$$

④ Calculate a_{14} :

The weights are:

$$w_{14}^{(2)} = 0.33$$

$$w_{24}^{(2)} = 0.02$$

Therefore:

$$z_{14} = x_1 w_{14}^{(2)} + x_2 w_{24}^{(2)}$$

$$= (0.5)(0.33) + (0.3)(0.02)$$

$$= 0.165 + 0.006$$

$$= 0.171$$

Apply Tanh:

$$a_{14} = \tanh(0.171)$$

$$a_{14} = 0.16935$$

All values of First Hidden Layer:

$$[a_{11}, a_{12}, a_{13}, a_{14}] = [0.20985, 0.35043, -0.04497, 0.16935]$$

Therefore:

$$Z_{22} = a_{11} w_{12}^{(2)} + a_{12} w_{22}^{(2)} + a_{13} w_{32}^{(2)} + a_{14} w_{42}^{(2)}$$

Substitute:

$$Z_{22} = (0.20484)(1.0) + (0.35049)(0.45) + (-0.04497)(-0.65) + (0.16935)(-0.57)$$

$$Z_{22} \approx 0.20484 + 0.15772 + 0.02923 - 0.09653$$

$$Z_{22} = 0.30026$$

Apply tanh:

$$a_{22} = \tanh(0.30026)$$

$$a_{22} \approx 0.29155$$

* ③ CALCULATE a_{23} :
The weights entering Tarc:

$$w_{13}^{(2)} = -0.8$$

$$w_{23}^{(2)} = 0.05$$

$$w_{33}^{(2)} = 0.33$$

$$w_{43}^{(2)} = 0.51$$

Therefore:

$$Z_{23} = a_{11} w_{13}^{(2)} + a_{12} w_{23}^{(2)} + a_{13} w_{33}^{(2)} + a_{14} w_{43}^{(2)}$$

Substitute:

$$Z_{23} = (0.20484)(-0.8) + (0.35049)(0.05) + (-0.04497)(\cancel{-0.33}) + (0.16935)(0.51)$$

$$Z_{23} \approx -0.16787 + 0.01752 + 0.01484 + 0.08637$$

$$Z_{23} \approx -0.04913$$

Apply tanh:

$$a_{23} = \tanh(-0.04913)$$

$$a_{23} \approx -0.04910$$

FINAL VALUES $\rightarrow$ 2nd Hidden Layer:

The values are: $[-0.06747, 0.29155, -0.04910]$

PROBLEM 02 PART A $\rightarrow$ output layer

Using the appropriate weights and activation function, compute the values of the nodes for the output layer.

Second layer values:

$$a_{11} \approx -0.06747$$

$$a_{22} \approx 0.29155$$

$$a_{33} \approx -0.04910$$

The output layer uses sigmoid.

Calculate O_1 :

The respective weights are:

$$w_{11}^{(3)} = 0.97$$

$$w_{21}^{(3)} = -0.01$$

$$w_{31}^{(3)} = 0.85$$

So the weighted sum entering O_1 is:

$$Z_1 = a_{11} w_{11}^{(3)} + a_{22} w_{21}^{(3)} + a_{33} w_{31}^{(3)}$$

substitute the values:

$$Z_1 = (-0.06747)(0.97) + (0.29155)(-0.01) + (-0.04910)(0.85)$$

calculate each term:

$$(-0.06747)(0.97) = -0.06542$$

$$(0.29155)(-0.01) = -0.00292$$

$$(-0.04910)(0.85) = -0.04174$$

Therefore

$$z_1 = -0.06542 - 0.00292 + 0.04174$$

$$z_1 \approx -0.11008$$

Now apply the sigmoid function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Therefore:

$$o_1 = \frac{1}{1 + e^{-(-0.11008)}}$$

$$o_1 = \frac{1}{1 + e^{0.11008}}$$

$$o_1 \approx 0.4725$$

Step 2: Calculate o_2

The respective weights are:

$$w_{12}^{(3)} = 0.30$$

$$w_{22}^{(3)} = -0.20$$

$$w_{32}^{(1)} = 0.76$$

Therefore:

$$z_2 = a_{21} w_{12}^{(3)} + a_{22} w_{22}^{(3)} + a_{23} w_{32}^{(1)}$$

substitute:

$$z_2 = (-0.06747)(0.30) + (0.29155)(-0.20) + (-0.04910)(0.76)$$

calculate:

$$(-0.06747)(0.30) = -0.02024$$

$$(0.29155)(-0.20) = -0.05831$$

$$(-0.04910)(0.76) = -0.03732$$

$$\text{Therefore: } z_2 = -0.02024 - 0.05831 - 0.03732$$

$$z_2 \approx -0.11587$$

$$\text{Apply sigmoid: } o_2 = \frac{1}{1 + e^{-(-0.11587)}} \rightarrow o_2 = \frac{1}{1 + e^{0.11587}}$$

$$\Rightarrow o_2 \approx 0.4711$$

④

BACKWARD PROPAGATION AND WEIGHT UPDATE

- ① By using partial derivatives, compute the derivative of the error w.r.t the weights $w_{31}^{[2]}$

$$\frac{\partial E}{\partial w_{31}^{[2]}}$$

The weight $w_{31}^{[2]}$ connects the third node of the hidden layer 2 to the first output node.
So the path is:

$$w_{31}^{[2]} \rightarrow z_1 \rightarrow o_1 \rightarrow E$$

where:

z_1 = input to output node 1

o_1 = output after sigmoid

E = error / cost

By chain rule:

$$\frac{\partial E}{\partial w_{31}^{[2]}} = \frac{\partial E}{\partial o_1} \cdot \frac{\partial o_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{31}^{[2]}}$$

→ This is the chain of derivatives we need, starting with:

$$\frac{\partial E}{\partial o_1}$$

Here, our job was simply to identify the chain of dependencies before calculating any derivative.

We want: $\frac{\partial E}{\partial w_{31}^{[2]}}$

→ The weight $w_{31}^{[2]}$ directly affects the input to the first output neuron, which then goes through the sigmoid to produce o_1 , which contributes to the Error E .

→ So the dependency is:

$$w_{31}^{[2]} \rightarrow z_1 \rightarrow o_1 \rightarrow E$$

Step 2:

Calculate $\frac{\partial E}{\partial o_1}$

From 1st step:

$$\frac{\partial E}{\partial w_{31}} = \frac{\partial E}{\partial o_1} \frac{\partial o_1}{\partial z_1} \frac{\partial z_1}{\partial w_{31}}$$

Calculating the first derivatives

$$\frac{\partial E}{\partial o_1}$$

→ We know that the cost function is Binary Cross Entropy and the actual value for the first output is

$$y_1 = 0.4$$

while the forward pass gave

$$o_1 = 0.4725$$

Binary Cross Entropy (BCE):

For one output, BCE is:

$$E_1 = -[y_1 \ln(o_1) + (1 - y_1) \ln(1 - o_1)]$$

We differentiate this w.r.t to o_1 :

$$\frac{\partial E}{\partial o_1} = -\frac{y_1}{o_1} + \frac{1 - y_1}{1 - o_1}$$

Now substitute

$$y_1 = 0.4 \quad \text{and} \quad o_1 = 0.4725$$

Therefore:

$$\begin{aligned} \frac{\partial E}{\partial o_1} &= -\frac{0.4}{0.4725} + \frac{1 - 0.4}{1 - 0.4725} \\ &= -\frac{0.4}{0.4725} + \frac{0.6}{0.5275} \end{aligned}$$

$$\approx -0.8466 + 1.1374$$

$$\boxed{\frac{\partial E}{\partial o_1} \approx 0.2908}$$

Step 3. Calculate $\frac{\partial o_1}{\partial z_1}$

In step 1, our chain-rule was:

$$\frac{\partial E}{\partial w_{s,1}^{(1)}} = \frac{\partial E}{\partial o_1} \frac{\partial o_1}{\partial z_1} \frac{\partial z_1}{\partial w_{s,1}^{(1)}}$$

We have calculated:

$$\frac{\partial E}{\partial o_1} \approx 0.2908$$

Now we need:

$$\frac{\partial o_1}{\partial z_1}$$

Now the output layer uses the sigmoid function

$$o_1 = \sigma(z_1)$$

where

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

The derivative of sigmoid is:

$$\frac{d\sigma(z)}{dz} = \sigma(z)(1-\sigma(z))$$

Since $o_1 = \sigma(z_1)$

we can write: $\frac{\partial o_1}{\partial z_1} = o_1(1-o_1)$

→ substitute the forward pass value:

we calculated:

$$o_1 \approx 0.4725$$

Therefore: $\frac{\partial o_1}{\partial z_1} = (0.4725)(1-0.4725)$

$$= (0.4725)(0.5275)$$

$$\boxed{\frac{\partial o_1}{\partial z_1} \approx 0.2492}$$

Step 4: Calculate $\frac{\partial z_1}{\partial w_{3,1}^{(3)}}$

From the forward pass, the input to the first node was:

$$z_1 = a_{21} w_{1,1}^{(3)} + a_{22} w_{2,1}^{(3)} + a_{23} w_{3,1}^{(3)}$$

We want the derivative with respect to $w_{3,1}^{(3)}$

Everything except the term containing $w_{3,1}^{(3)}$ is constant with respect to $w_{3,1}^{(3)}$

Therefore:

$$\frac{\partial z_1}{\partial w_{3,1}^{(3)}} = a_{23}$$

From ~~our~~ forward pass:

$$a_{23} = -0.04910$$

Therefore:

$$\boxed{\frac{\partial z_1}{\partial w_{3,1}^{(3)}} \approx -0.04910}$$

FINAL DERIVATIVE:

$$\frac{\partial E}{\partial w_{3,1}^{(3)}} = \frac{\partial E}{\partial o_1} \frac{\partial o_1}{\partial z_1} \frac{\partial z_1}{\partial w_{3,1}^{(3)}}$$

Substitute:

$$= (0.2908) (0.2492) (-0.04910)$$

$$\boxed{\frac{\partial E}{\partial w_{3,1}^{(3)}} \approx -0.00356}$$

5

BACKWARD PROPAGATION QUESTION 2

Now we have to calculate:

$$\frac{\partial E}{\partial w_{23}^{(2)}}$$

STEP 1.

Identify the complete chain:

Unlike $w_{11}^{(1)}$, this weight is in Hidden Layer 2, so it affects both the output nodes.

The path is:

$$w_{23}^{(2)} \rightarrow z_{23} \rightarrow a_{23} \rightarrow \begin{cases} z_1 \rightarrow o_1 \rightarrow E \\ z_2 \rightarrow o_2 \rightarrow E \end{cases}$$

Using chain rules:

$$\frac{\partial E}{\partial w_{23}^{(2)}} = \frac{\partial E}{\partial o_1} \frac{\partial o_1}{\partial z_1} \frac{\partial z_1}{\partial a_{23}} \frac{\partial a_{23}}{\partial z_{23}} + \frac{\partial E}{\partial o_2} \frac{\partial o_2}{\partial z_2} \frac{\partial z_2}{\partial a_{23}} \frac{\partial a_{23}}{\partial z_{23}} \frac{\partial z_{23}}{\partial w_{23}^{(2)}}$$

Now we'll calculate the first required component: $\frac{\partial E}{\partial o_1}$

We already have this from B1:

$$\frac{\partial E}{\partial o_1} = 0.2908$$

We also need the second output's error derivative:

$$\frac{\partial E}{\partial o_2}$$

Using (BCE):

$$\frac{\partial E}{\partial o_2} = -\frac{y_2}{o_2} + \frac{1-y_2}{1-o_2}$$

$$\rightarrow y_2 = 0.45$$

and from forward pass:

$$o_2 = 0.4711$$

$$\text{Therefore } \frac{\partial E}{\partial o_2} = -\frac{0.45}{0.4711} + \frac{0.55}{1-0.4711}$$

$$= -\frac{0.45}{0.4711} + \frac{0.55}{0.5289}$$

$$\approx -0.9552 + 1.0399 \rightarrow \frac{\partial E}{\partial o_2} \approx 0.0847$$

At this point we have:

$$\frac{\partial E}{\partial o_1} \approx 0.2908$$

$$\frac{\partial E}{\partial o_2} \approx 0.0847$$

Now we'll calculate the sigmoid derivatives
 $\frac{\partial o_1}{\partial z_1}$ and $\frac{\partial o_2}{\partial z_2}$

step 3: Sigmoid derivatives:

We need:

$$\frac{\partial o_1}{\partial z_1} \text{ and } \frac{\partial o_2}{\partial z_2}$$

For sigmoid:

$$\frac{\partial o}{\partial z} = o(1-o)$$

For o_1 , we have:

$$o_1 \approx 0.4725$$

$$\begin{aligned} \text{So, } \frac{\partial o_1}{\partial z_1} &= (0.4725)(1-0.4725) \\ &= (0.4725)(0.5275) \\ \frac{\partial o_1}{\partial z_1} &\approx 0.2492 \end{aligned}$$

For o_2 , we have:

$$o_2 \approx 0.4711$$

Therefore,

$$\begin{aligned} \frac{\partial o_2}{\partial z_2} &= (0.4711)(1-0.4711) \\ &= (0.4711)(0.5289) \\ \frac{\partial o_2}{\partial z_2} &\approx 0.2492 \end{aligned}$$

Now we have:

$$\frac{\partial E}{\partial o_1} \approx 0.2908$$

$$\frac{\partial o_1}{\partial z_1}$$

$$\frac{\partial E}{\partial o_2} \approx 0.0847$$

$$\frac{\partial o_2}{\partial z_2}$$

$$\frac{\partial o_1}{\partial z_1} \approx 0.2492$$

$$\frac{\partial o_2}{\partial z_2}$$

$$\frac{\partial o_2}{\partial z_2} \approx 0.2492$$

$$\frac{\partial o_2}{\partial z_2}$$

Next, we need to calculate how a_{23} affects each output. i.e.,

$$\frac{\partial z_1}{\partial a_{23}} \quad \text{and} \quad \frac{\partial z_2}{\partial a_{23}}$$

STEP 4: How a_{23} affects the outputs:

We need:

$$\frac{\partial z_1}{\partial a_{23}} \quad \text{and} \quad \frac{\partial z_2}{\partial a_{23}}$$

From the output-layer eq:-

$$z_1 = a_{21}w_{11}^{(3)} + a_{22}w_{21}^{(3)} + a_{23}w_{31}^{(3)}$$

Therefore:

$$\frac{\partial z_1}{\partial a_{23}} = w_{31}^{(3)}$$

From the given weights:

$$w_{31}^{(3)} = 0.85$$

So,

$$\boxed{\frac{\partial z_1}{\partial a_{23}} = 0.85}$$

For the second output:

$$z_2 = a_{21}w_{12}^{(3)} + a_{22}w_{22}^{(3)} + a_{23}w_{32}^{(3)}$$

Therefore:

$$\frac{\partial z_2}{\partial a_{23}} = w_{32}^{(3)}$$

Given:

$$w_{32}^{(3)} = 0.76 \rightarrow \text{So}$$

$$\boxed{\frac{\partial z_2}{\partial a_{23}} = 0.76}$$

STEP 5: Calculate $\frac{\partial a_{23}}{\partial z_{23}}$

The second hidden layer uses the tanh activation

So:

$$a_{23} = \tanh(z_{23})$$

The derivative of tanh is: $\frac{d}{dx} \tanh(x) = 1 - \tanh^2(x)$

Since: $a_{23} = \tanh(z_{23})$

We can write:

$$\frac{\partial z_{23}}{\partial z_{23}} = 1 - a_{23}^2$$

From forward pass:

$$a_{23} = -0.04910$$

Therefore:

$$\frac{\partial z_{23}}{\partial z_{23}} = 1 - (-0.04910)^2$$
$$= 1 - 0.0024108$$

$$\boxed{\frac{\partial z_{23}}{\partial z_{23}} \approx 0.99759}$$

Step 6: Calculate:

$$\frac{\partial z_{23}}{\partial w_{23}^{(2)}}$$

From the second hidden Layer:

$$z_{23} = a_{11}^{(2)} w_{13}^{(2)} + a_{12}^{(2)} w_{23}^{(2)} + a_{13}^{(2)} w_{33}^{(2)} + a_{14}^{(2)} w_{43}^{(2)}$$

We need:

$$\frac{\partial z_{23}}{\partial w_{23}^{(2)}}$$

only the term: $a_{12} w_{23}^{(2)}$ contains $w_{23}^{(2)}$

Therefore

$$\boxed{\frac{\partial z_{23}}{\partial w_{23}^{(2)}} = a_{12}}$$

From forward pass $a_{12} \approx 0.35049$

Therefore:

$$\boxed{\frac{\partial z_{23}}{\partial w_{23}^{(2)}} \approx 0.35049}$$

$\Rightarrow$ substitute all the values we calculated:

$$= (0.2968)(0.2492)(0.85)(0.99759)(0.35049) + (0.0847)(0.2492)(0.76)(0.99759)(0.35049)$$

First path ≈ 0.02170 , second path ≈ 0.00545

Therefore: $\frac{\partial E}{\partial w_{23}^{(2)}} = 0.02170 + 0.00545 \Rightarrow \boxed{\frac{\partial E}{\partial w_{23}^{(2)}} \approx 0.02715}$

⑥

BACKWARD PROPAGATION QUESTION 02

(B3)

Step 1: Identify the chain

We need to calculate:

$$\frac{\partial E}{\partial w_{12}^{(1)}}$$

→ Here, $w_{12}^{(1)}$ is in the first hidden layer, so it affects a_{12} , which then affects both nodes in the second hidden layer, and from there both output nodes.

→ There are 2 main paths, each reaching the error:

$$w_{12}^{(1)} \rightarrow z_{12} \rightarrow a_{12} \rightarrow \begin{cases} z_{21} \rightarrow a_{21} \rightarrow o_1 \rightarrow E \\ z_{22} \rightarrow a_{22} \rightarrow o_2 \rightarrow E \\ z_{23} \rightarrow a_{23} \rightarrow o_1 \rightarrow E \\ z_{23} \rightarrow a_{23} \rightarrow o_2 \rightarrow E \end{cases}$$

a_{12} influences all 3 second-hidden-layer nodes, and each of these can influence the output error.

→ Using the chain rule, we can write:

$$\frac{\partial E}{\partial w_{12}^{(1)}} = \frac{\partial E}{\partial a_{12}} \frac{\partial a_{12}}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{12}^{(1)}}$$

So our calculation will be:

$$w_{12}^{(1)} \rightarrow z_{12} \rightarrow a_{12} \rightarrow (a_{12}, a_{22}, a_{23}) \rightarrow (o_1, o_2) \rightarrow E$$

Step 2: Contributing through a_{21} :

$$a_{12} \rightarrow a_{21} \rightarrow o_1, o_2 \rightarrow E$$

calculate how a_{12} affects the error:

$$\frac{\partial E}{\partial a_{21}} = \frac{\partial E}{\partial o_1} \frac{\partial o_1}{\partial z_{21}} + \frac{\partial E}{\partial o_2} \frac{\partial o_2}{\partial z_{21}} \frac{\partial z_{21}}{\partial a_{21}}$$

We know:

$$\frac{\partial E}{\partial o_1} = 0.2908$$

$$\frac{\partial o_1}{\partial z_{21}}$$

$$\frac{\partial o_1}{\partial z_{21}} = 0.2492$$

$$\frac{\partial z_{21}}{\partial a_{21}}$$

$$\frac{\partial E}{\partial a_2} = 0.0847$$

$$\frac{\partial a_2}{\partial z_2}$$

$$\frac{\partial a_2}{\partial z_2} = 0.2492$$

$$\frac{\partial z_2}{\partial a_1}$$

From the output-layer weights:

$$\frac{\partial z_1}{\partial a_1} = w_{11}^{(3)} = 0.97$$

$$\frac{\partial z_2}{\partial a_1} = w_{12}^{(3)} = 0.30$$

Therefore,

$$\frac{\partial E}{\partial a_1} = (0.2908)(0.2492)(0.97) + (0.0847)(0.2492)(0.30)$$

First path: $= 0.07027$

second path: $= 0.00633$

Therefore,

$$\frac{\partial E}{\partial a_1} \approx 0.07660$$

Step 3: Contribution through a_{22} :

$$\frac{\partial E}{\partial a_{22}} = \frac{\partial E}{\partial a_1} \frac{\partial a_1}{\partial z_1} \frac{\partial z_1}{\partial a_{22}} + \frac{\partial E}{\partial a_2} \frac{\partial a_2}{\partial z_2} \frac{\partial z_2}{\partial a_{22}}$$

From the output weights:

$$\frac{\partial z_1}{\partial a_{22}} = w_{21}^{(3)} = -0.01$$

$$\frac{\partial z_2}{\partial a_{22}} = w_{22}^{(3)} = -0.20$$

substitute:

$$\frac{\partial E}{\partial a_{22}} = (0.2908)(0.2492)(-0.01) + (0.0847)(0.2492)(-0.20)$$

1st path: $= -0.00072$

2nd path: $= -0.00422$

Therefore,

$$\frac{\partial E}{\partial a_{12}} = -0.00494$$

Step 4: Contribution through a_{23}

Now calculate:

$$\frac{\partial E}{\partial a_{23}}$$

$$\frac{\partial E}{\partial a_{23}} = \frac{\partial E}{\partial o_1} \frac{\partial o_1}{\partial z_1} \frac{\partial z_1}{\partial a_{23}} + \frac{\partial E}{\partial o_2} \frac{\partial o_2}{\partial z_2} \frac{\partial z_2}{\partial a_{23}}$$

We have:

$$\frac{\partial E}{\partial o_1} = 0.2908$$

$$\frac{\partial o_1}{\partial z_1}$$

$$\frac{\partial o_1}{\partial z_1} = 0.2492$$

$$\frac{\partial z_1}{\partial a_{23}}$$

$$\frac{\partial E}{\partial o_2} = 0.0847$$

$$\frac{\partial o_2}{\partial z_2}$$

$$\frac{\partial o_2}{\partial z_2} = 0.2492$$

$$\frac{\partial z_2}{\partial a_{23}}$$

→ From the output weights:

$$\frac{\partial z_1}{\partial a_{23}} = w_{21}^{(2)} = 0.85$$

$$\frac{\partial z_2}{\partial a_{23}}$$

$$\frac{\partial z_2}{\partial a_{23}} = w_{22}^{(3)} = 0.76$$

$$\frac{\partial z_2}{\partial a_{23}}$$

Therefore:

$$\frac{\partial E}{\partial a_{23}} = (0.2908)(0.2492)(0.85) + (0.0847)(0.2492)(0.76)$$

$$1^{\text{st}} \text{ path} \rightarrow 0.06163$$

$$2^{\text{nd}} \text{ path} \rightarrow 0.01605$$

Therefore: $\frac{\partial E}{\partial a_{23}} = 0.07768$

Step 5: Calculate $\frac{\partial E}{\partial a_{12}}$

We now move backward from a_{21}, a_{22}, a_{23} to a_{12}

$$\frac{\partial E}{\partial a_{12}} = \frac{\partial E}{\partial a_{21}} \frac{\partial a_{21}}{\partial z_{11}} \frac{\partial z_{11}}{\partial a_{12}} + \frac{\partial E}{\partial a_{22}} \frac{\partial a_{22}}{\partial z_{12}} \frac{\partial z_{12}}{\partial a_{12}} + \frac{\partial E}{\partial a_{23}} \frac{\partial a_{23}}{\partial z_{13}} \frac{\partial z_{13}}{\partial a_{12}}$$

We already have

$$\frac{\partial E}{\partial a_{21}} = 0.07660$$

$$\frac{\partial E}{\partial a_{22}} = 0.00494$$

$$\frac{\partial E}{\partial a_{23}} = 0.07768$$

For tanh:

$$\frac{\partial a}{\partial z} = 1 - a^2$$

So:

$$\frac{\partial a_{21}}{\partial z_{21}} = 1 - (-0.06147)^2 \approx 0.99545$$

$$\frac{\partial a_{22}}{\partial z_{22}} = 1 - (0.29155)^2 \approx 0.91500$$

$$\frac{\partial a_{23}}{\partial z_{23}} = 1 - (-0.04910)^2 \approx 0.99759$$

Weights from a_{12}

From the network:

$$\frac{\partial z_{21}}{\partial a_{12}} = w_{21}^{(2)} = 0.23$$

$$\frac{\partial z_{22}}{\partial a_{12}} = w_{22}^{(2)} = 0.45$$

$$\frac{\partial z_{23}}{\partial a_{12}} = w_{23}^{(2)} = 0.05$$

Now substitute:

$$\frac{\partial E}{\partial a_{12}} = (0.07660)(0.99545)(0.23) + (-0.00494)(0.91500)(0.45) +$$

$$(0.07768)(0.99759)(0.05)$$

Calculate:

$$\approx 0.01754$$

$$\approx -0.00203$$

$$\approx 0.00387$$

Therefore:

$$\boxed{\frac{\partial E}{\partial a_{12}} \approx 0.019388}$$

⑦

Step 6: Calculate $\frac{\partial a_{12}}{\partial z_{12}}$

We have:

$$a_{12} = \tanh(z_{12})$$

For $\tanh$:

$$\frac{\partial a_{12}}{\partial z_{12}} = 1 - a_{12}^2$$

We know:

$$a_{12} = 0.35049$$

Therefore:

$$\frac{\partial a_{12}}{\partial z_{12}} = 1 - (0.35049)^2$$

$$\approx 1 - 0.12284$$

$$\boxed{\frac{\partial a_{12}}{\partial z_{12}} \approx 0.87716}$$

Step 7: Calculate $\frac{\partial z_{12}}{\partial w_{12}^{(1)}}$

First hidden-layer node is:

$$z_{12} = w_{11}^{(1)} + w_{12}^{(1)}$$

Therefore:

$$\frac{\partial z_{12}}{\partial w_{12}^{(1)}} = 1$$

So:

$$\boxed{\frac{\partial z_{12}}{\partial w_{12}^{(1)}} = 0.5}$$

Step 8: Final derivative:

Now combine everything

$$\frac{\partial E}{\partial w_{12}^{(1)}} = \frac{\partial E}{\partial a_{12}} \frac{\partial a_{12}}{\partial z_{12}} \frac{\partial z_{12}}{\partial w_{12}^{(1)}}$$

substitute:

$$= (0.01938) (0.87716) (0.5)$$

$$\boxed{\frac{\partial E}{\partial w_{12}^{(1)}} \approx 0.00850}$$

→ Final answer for B3

BACKWARD PROPAGATION QUESTION 4

WEIGHT UPDATE

The Learning rate is 0.01 and update:

$$w_{21}^{(3)}, w_{23}^{(2)}, w_{12}^{(1)}$$

using the derivatives that we calculated.
→ Gradient descent formula is:

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial E}{\partial w}$$

where:

$$\eta = 0.01$$

1. Update $w_{21}^{(3)}$:

Original value:

$$w_{21}^{(3)} = 0.85$$

Derivative from B1:

$$\frac{\partial E}{\partial w_{21}^{(3)}} = -0.00356$$

Therefore:

$$\begin{aligned} w_{21}^{(3)\text{new}} &= 0.85 - (0.01)(-0.00356) \\ &= 0.85 + 0.000356 \end{aligned}$$

$$w_{21}^{(3)\text{new}} = 0.850356$$

2) Update $w_{23}^{(2)}$:

original value:

$$w_{23}^{(2)} = 0.05$$

Derivative from B2:

$$\frac{\partial E}{\partial w_{23}^{(2)}} = 0.02715$$

So,

$$\begin{aligned} w_{23}^{(2)\text{new}} &= 0.05 - (0.01)(0.02715) \\ &= 0.05 - 0.0002715 \end{aligned}$$

$$w_{23}^{(2)\text{new}} = 0.0497285$$

2) Update $w_{12}^{(1)}$:

Original value:

$$w_{12}^{(1)} = 0.75$$

Derivative from B3:

$$\frac{\partial E}{\partial w_{12}^{(1)}} = 0.00850$$

So,

$$w_{12}^{(1)\text{new}} = 0.75 - (0.01) (0.00850) \\ = 0.75 - 0.000085$$

$$w_{12}^{(1)\text{new}} = 0.749915$$

Final Answers $\rightarrow$ B4

Weight	old value	Derivative	New value
$w_{31}^{(2)}$	0.85	-0.00356	0.8500356
$w_{23}^{(2)}$	0.05	0.02715	0.0497285
$w_{12}^{(1)}$	0.75	0.00850	0.749915